

US Consumer Price Index CPI

Forecasting Techniques Project

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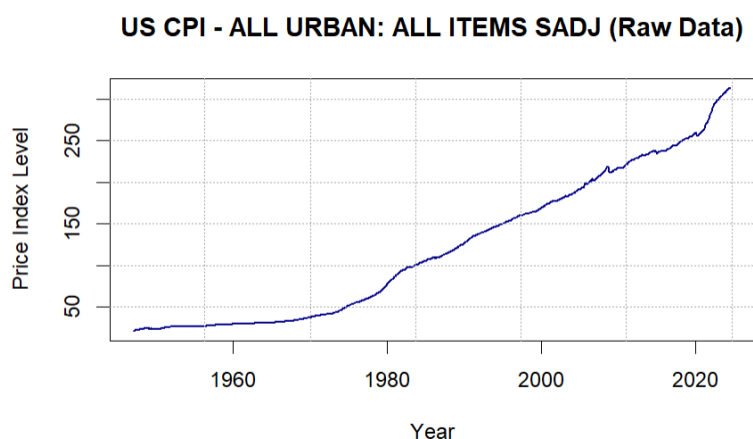
2nd Trimester - Course 2025-2026

INTRODUCTION

The US CPI – All Urban: All Items, Seasonally Adjusted is the most widely followed measure of consumer inflation in the United States. Formally produced by the U.S. Bureau of Labor Statistics, the Consumer Price Index (CPI) tracks the average change over time in the prices paid by urban consumers for a broad basket of goods and services. The “All Urban” population (CPI-U) represents roughly 93% of the U.S. population, covering residents of urban and metropolitan areas, while “All Items” refers to the comprehensive basket including food, energy, housing, transportation, medical care, recreation, education, and other categories. The “seasonally adjusted” (SADJ) designation means the index has been statistically adjusted to remove predictable seasonal patterns, such as higher gasoline demand in summer or holiday-related price fluctuations, so that month to month movements better reflect underlying inflation dynamics.

Conceptually, CPI is a cost of living indicator. It measures price changes from the consumer’s perspective rather than the producer’s, making it a central benchmark for assessing purchasing power, wage adjustments, Social Security cost of living adjustments, and inflation indexed contracts. Policymakers, especially the Federal Reserve, monitor CPI closely when evaluating inflation conditions, even though the Fed formally targets inflation using the Personal Consumption Expenditures (PCE) price index. Financial markets treat CPI releases as major macroeconomic events, often generating immediate reactions in bond yields, equity prices, and exchange rates.

History of the CPI:



Historically, CPI has exhibited distinct inflation regimes shaped by economic structure and macroeconomic shocks. During the 1970s, the index rose rapidly amid oil supply disruptions such as the 1973 oil crisis and the 1979 energy crisis, combined with accommodative monetary policy and wage-price spirals. Inflation peaked above 13% year over year in 1980. The aggressive tightening campaign under Federal Reserve Chair Paul Volcker in the early 1980s successfully reduced inflation, ushering in the “Great Moderation,” a period from the mid 1980s through the mid-2000s characterized by relatively low and stable CPI growth, typically around 2–3% annually.

The 2008 financial crisis marked another turning point. During the Global Financial Crisis, CPI inflation briefly turned negative as energy prices collapsed and aggregate demand weakened. This period highlighted CPI’s sensitivity to energy price swings and macroeconomic contraction. In the

following decade, inflation remained subdued despite economic recovery, reflecting globalization, technological change, anchored inflation expectations, and labor market slack.

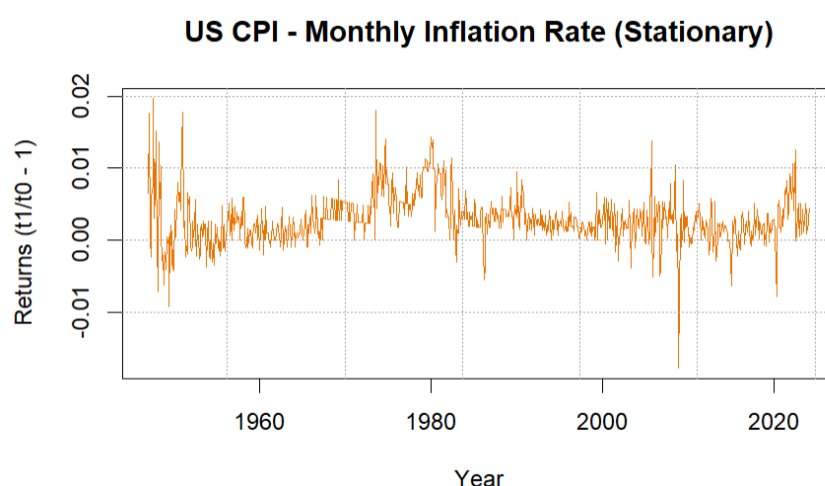
A dramatic shift occurred in the aftermath of the COVID-19 pandemic. Supply chain disruptions, expansive fiscal stimulus, rapid demand recovery, and tight labor markets led to a surge in CPI inflation beginning in 2021. By mid 2022, year over year CPI inflation exceeded 9%, the highest rate in four decades. Energy and food prices initially drove the surge, but price pressures broadened into housing and core services categories, reflecting more persistent inflation dynamics. In response, the Federal Reserve implemented one of the fastest monetary tightening cycles in modern history, sharply raising interest rates to restore price stability.

The index:

From a statistical perspective, CPI is an index number constructed using a modified Laspeyres formula, with expenditure weights derived from consumer surveys. Housing rent carries the largest weight, making shelter a dominant contributor to overall CPI movements. Energy and food components, while volatile, can significantly influence short term fluctuations. Because CPI is seasonally adjusted, its month to month movements are particularly useful for identifying turning points and underlying momentum, though year over year rates remain the most commonly cited measure.

Several key characteristics define the behavior of CPI. First, it tends to display persistence: inflation shocks often propagate over multiple months due to contract structures, wage adjustments, and expectation formation. Second, it is sensitive to commodity price cycles, especially oil. Third, it reflects both demand side forces (such as fiscal stimulus or monetary easing) and supply-side constraints (such as supply chain disruptions or geopolitical shocks).

Stationarity of the index:



Visually, this raw index is a prime example of a non-stationary variable, as it lacks a constant mean and shows a clear, powerful trend that makes it difficult to predict in its original form. To uncover the actual volatility and short-term behavior of inflation, the data must be transformed by taking first differences, which shifts the focus from the absolute price level to the Monthly Inflation Rate. In

concrete we used the following formula for arithmetic returns: $r_t = \left(\frac{CPI_t}{CPI_{t-1}} - 1 \right)$ This transformed "Returns" series represents the percentage change from one month to the next. Unlike the raw index, this stationary version of the variable oscillates around a stable center, revealing the real-time "shocks" and adjustments that occur within the US economy, allowing for a deeper statistical identification of the patterns that drive modern inflation.

It can be clearly seen that the mean is different from 0. In fact it is equal to 0.28925% in monthly terms and approximately 3.5% annually. This aligns perfectly with the historical records, where inflation typically targets the 2-3% range but can lean slightly higher during certain cycles.

For the purposes of this project, from now on we will use the stationary series for the analysis.

IDENTIFICATION

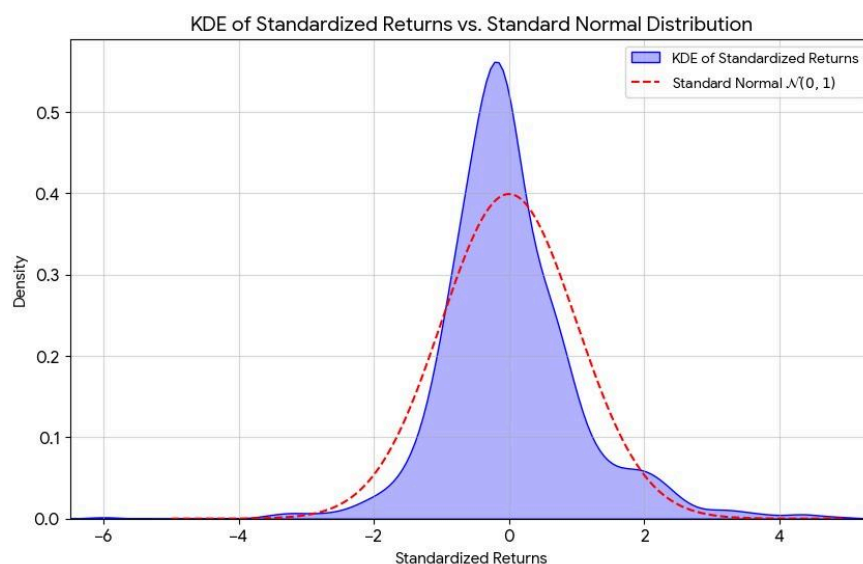
In this phase of the study, the goal is to transition from raw data observation to formal statistical modeling by identifying which mathematical structure best represents the inflation dynamics observed in the US CPI.

Kernel Density Estimator (KDE)

To give it more shape and in order to check distributional assumptions with respect to normality, we will plot the Kernel Density Estimator.

It is like a histogram but smoother. It allows data to “show its nature” and reveal its true underlying shape.

KDE also helps us identify Kurtosis and spot asymmetry.



Visually, the KDE does not match the bell shape of a Standard Normal distribution.

We can see a sharper central peak and thicker tails, which indicates that returns cluster very tightly around the mean, but when deviations happen, they are much larger than a normal distribution would predict.

The slight positive skewness tells us that extreme positive returns happen slightly more often than negative ones.

Lastly, vast majority of data points are squeezed into this narrow band, reinforcing a steady, low volatility mixed with occasional large jumps.

The Ljung-Box (LBO) Test:

Before committing to any specific model, we must determine if the "Returns" series contains enough predictable information to justify a complex ARMA structure. The Ljung-Box test is designed to examine whether the autocorrelations of a time series are different from zero; essentially, it tests if the data is White Noise or if it possesses memory. In particular, it tests the Null Hypothesis of $H_0 = \rho_1 = 0, \rho_2 = 0, \dots, \rho_m = 0$.

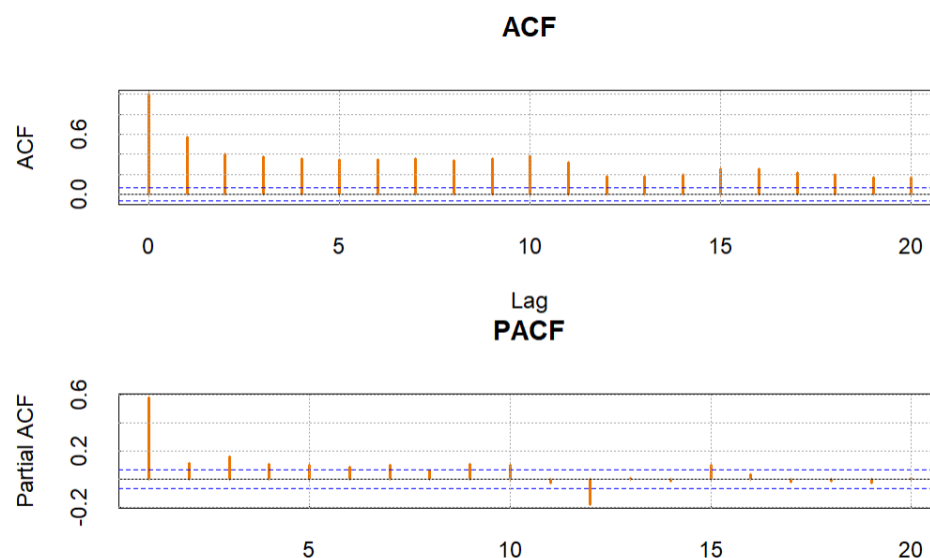
Box-Ljung test

```
data: y
X-squared = 1921.3, df = 22, p-value < 2.2e-16
```

Our results show an X-squared value of 1921.3 and a tiny p-value. Because this p-value is effectively zero, we reject the null hypothesis that the series is random. This confirms that the US CPI returns have a deep internal structure, proving that past inflation rates significantly influence current ones and that a forecasting model is viable.

ACF and PACF:

The ACF measures the direct correlation between observations at different distances, while the PACF measures that same correlation but removes the influence of the intervening lags.



In our analysis of 20 lags, the ACF shows a significant initial spike followed by a very slow, geometric decay. This pattern is a classic signature of an Autoregressive (AR) process.

Simultaneously, the PACF shows a sharp, significant spike at lag 1 and a smaller but notable one at lag 2, before dropping into the "noise" zone. This suggests that a first or second order AR model is a strong starting point, as the memory of the series seems strongest within the first two months.

Log-Likelihood AIC and BIC:

Following the identification of the underlying patterns, the project moves into the critical stage of model selection, where we determine which mathematical structure best captures the essence of US inflation.

The ACF shows a slow, geometric decay, while the PACF exhibits sharp, significant spikes in the first two months before cutting off. This visual evidence suggests that the series is driven by its own immediate past, leading us to propose five candidate models: the AR(1), AR(2), MA(1), ARMA(1,1), and ARMA(2,1).

We limit ourselves to these five options (we don't include 3 lags for example) to follow the principle of parsimony, which dictates that a model should be as simple as possible to avoid "overfitting," where a model becomes so complex that it starts memorizing random noise instead of learning real trends.

To evaluate and rank the five candidate models, we rely on the Log-Likelihood, the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC). The Log-Likelihood measures how well a model explains the observed data, with higher values indicating a superior fit. However, because adding more parameters always increases the likelihood, we also employ the AIC and BIC as scoring systems. These criteria reward a model for a high Log-Likelihood but penalize it for every extra parameter added, with the BIC providing a much stricter penalty for complexity to ensure the model remains simple and efficient.

	AR(1)	AR(2)	MA(1)	ARMA(1,1)	ARMA(2,1)
LogLik	4144.82693	4151.21710	4092.22485	4161.86519	4182.13118
AIC	-8.90931	-8.92090	-8.79618	-8.94380	-8.98523
BIC	-8.89891	-8.90530	-8.78578	-8.92820	-8.96443

In our selection process, we seek the model that achieves the highest Log-Likelihood while maintaining the lowest AIC and BIC values, representing the optimal balance between explanatory power and parsimony. Upon reviewing the results, the ARMA(2,1) model emerges as the superior choice, as it provides the highest Log-Likelihood of 4182.13 and achieves the lowest AIC of -8.985 and BIC of -8.964.

ESTIMATION

The primary objective of estimation is to determine the precise values of the model's parameters: the autoregressive coefficients (ϕ) and the moving average coefficient (θ). We want to find the exact numerical weight that the inflation from one and two months ago carries over to the present. To do this, we use Maximum Likelihood Estimation (MLE). This statistical method works by searching for the specific parameter values that maximize the probability of observing our actual historical US CPI data.

The results of the MLE for our ARMA(2,1) model provide the following coefficients:

```
[1] "--- Estimated Parameters ---"
      AR(1) AR(2) MA(1) ARMA(1,1) ARMA(2,1)
phi_0    0.00  0.00  0.00      0.00      0.00
phi_1    0.57  0.51      0.89      1.26
phi_2      0.12      -0.29
theta_1      0.47    -0.56    -0.85
```

The high positive AR(1) of 1.26 indicates a very strong momentum; if inflation was high last month, it exerts a powerful upward pressure on this month's rate. However, the negative AR(2) of -0.29 acts as a vital "brake" or stabilizer, suggesting a natural correction that prevents inflation from growing indefinitely.

Together, these parameters confirm that US inflation is a process with deep memory and significant inertia, but one that ultimately possesses an internal structure that prevents it from spiraling. With these exact coordinates now finalized, we have the mathematical equation necessary to project the path of inflation into the coming months.

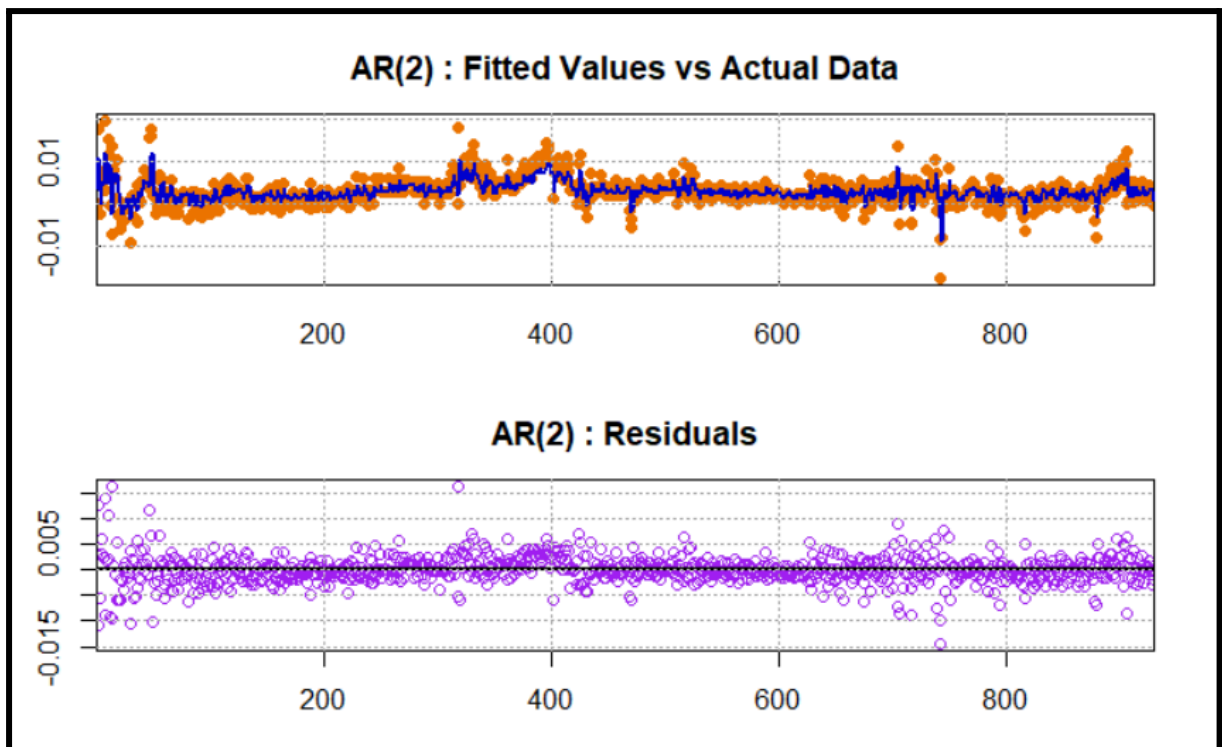
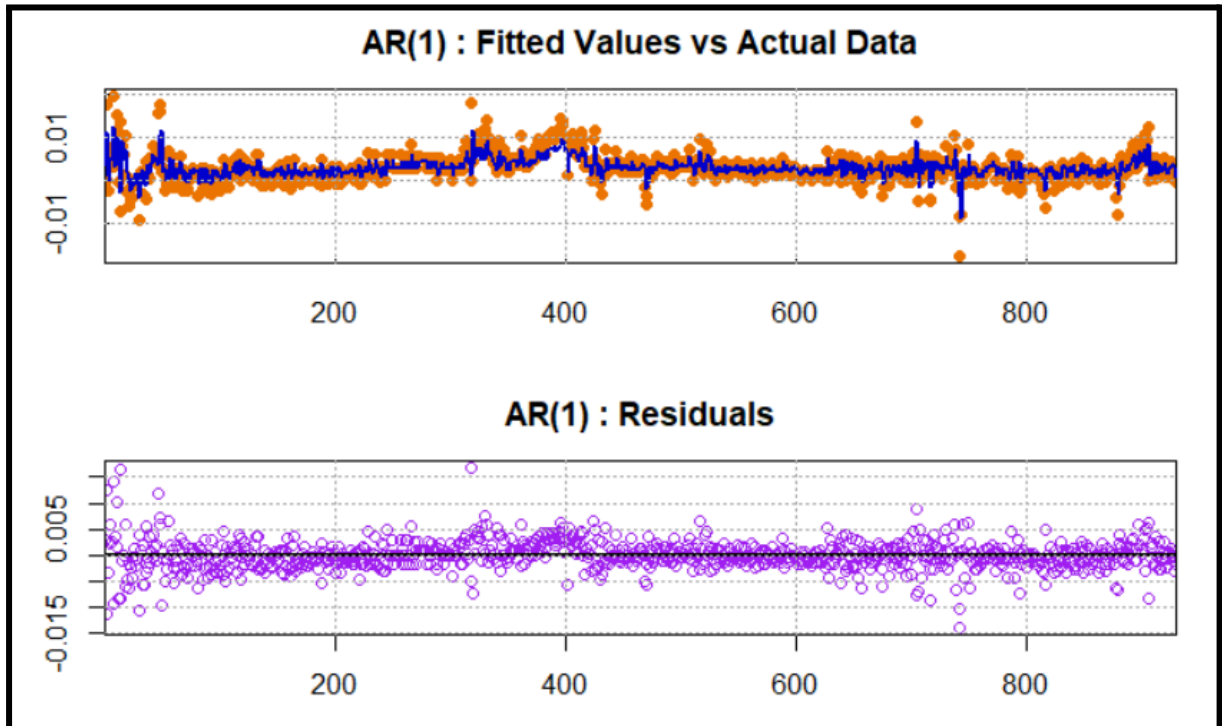
DIAGNOSTIC CHECKS

Model Innovations $y_t - \hat{y}_t$

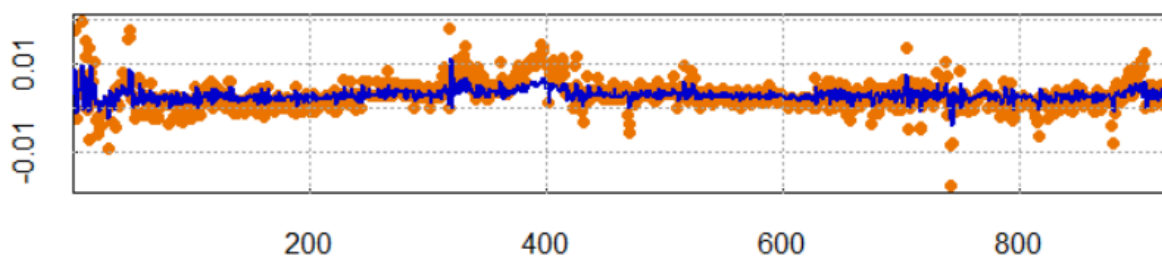
The "innovations," also known as residuals, represent the portion of the data that our model was unable to explain. Formally, they are defined as the difference between the actual observed CPI return values (y_t) and the values predicted by our fitted ARMA model ($\hat{y}_t$). In other words, they capture the "surprise" component of each month's inflation, the part that no mathematical model could have seen coming.

In a well-specified model, innovations should behave as White Noise: purely random fluctuations with no discernible patterns, trends, or clusters over time. If a systematic pattern were visible in the residuals, it would be a clear signal that the model is failing to capture important information still present in the data, information that, if incorporated, could improve our forecasts.

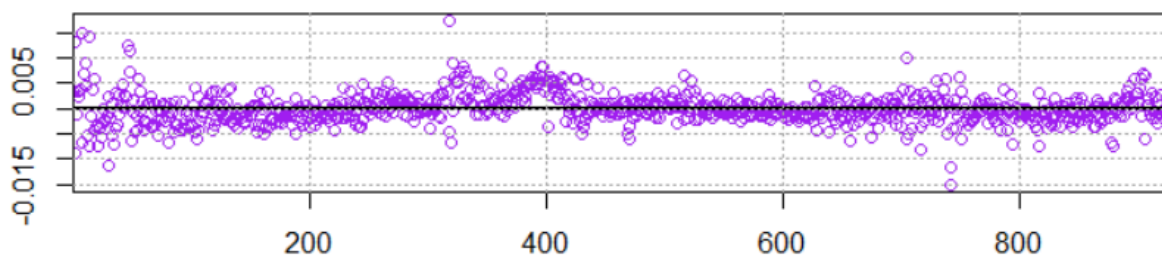
To assess this, we plot the fitted values against the actual data and examine the residuals over time for each of our candidate models: AR(1), AR(2), MA(1), ARMA(1,1), and our selected ARMA(2,1).



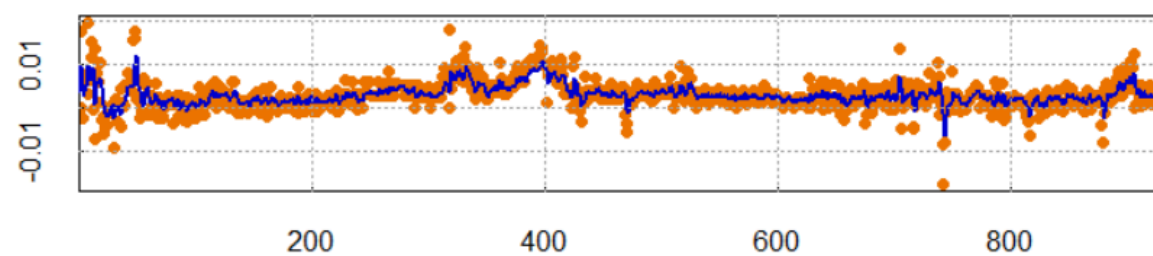
MA(1) : Fitted Values vs Actual Data



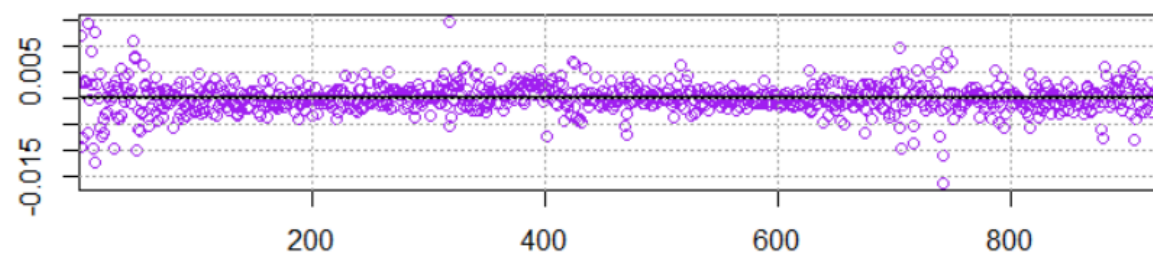
MA(1) : Residuals

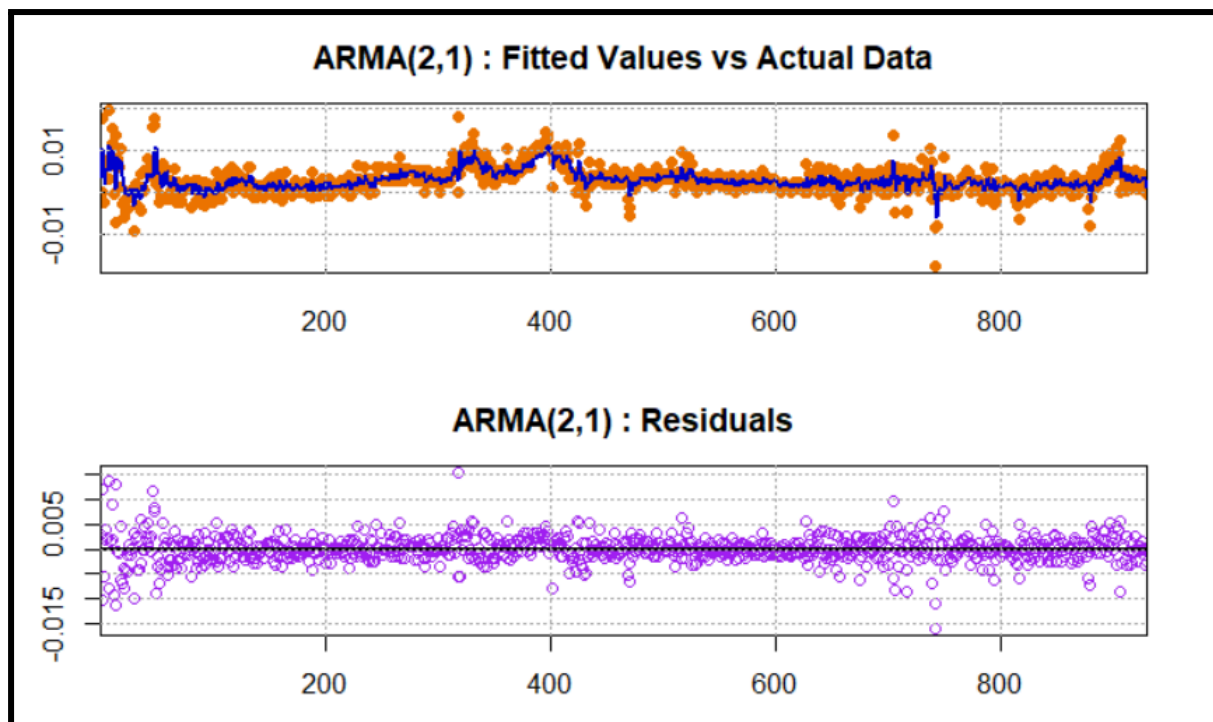


ARMA(1,1) : Fitted Values vs Actual Data



ARMA(1,1) : Residuals





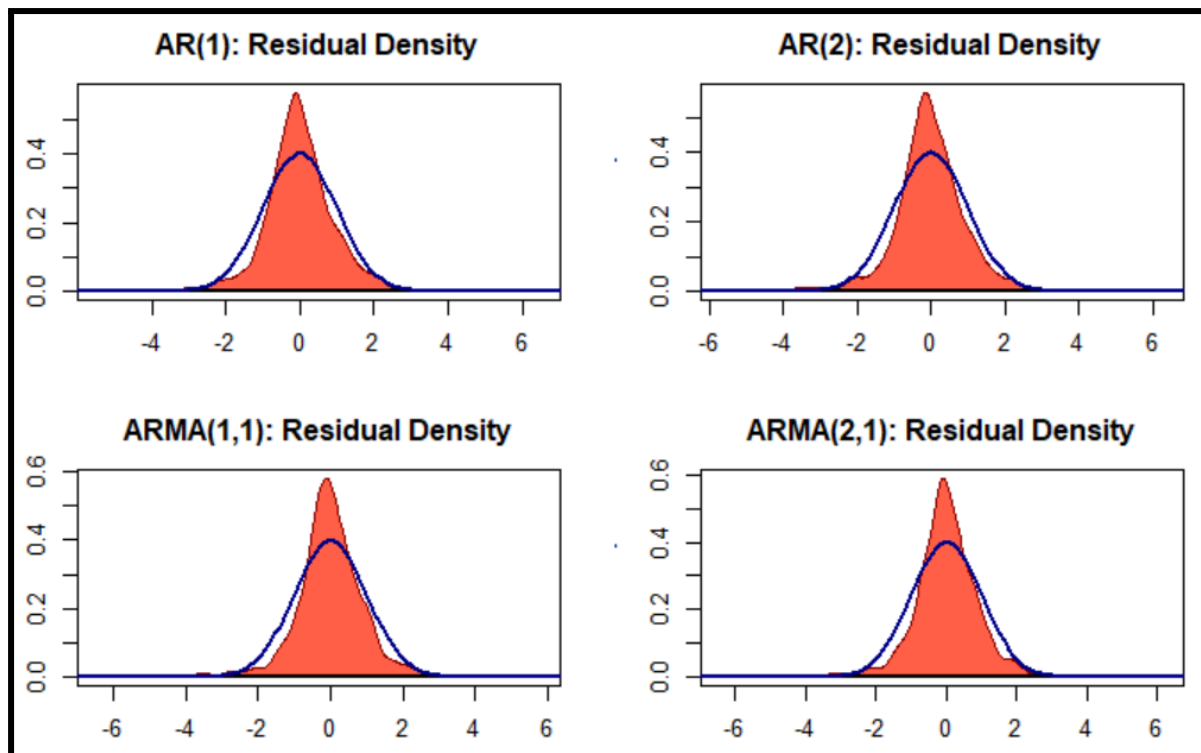
Inspecting the residual plots across all five models, we observe that in every case the residuals oscillate randomly around zero, with no visible upward or downward drift and no obvious clustering in any particular time period. Crucially, the ARMA(2,1) model produces the most well-behaved residuals among all candidates. While simpler models such as AR(1) and MA(1) already show broadly random behavior, the ARMA(2,1) residuals are noticeably tighter and more evenly distributed around zero throughout the entire sample period, from the early 1960s through to the post-pandemic inflation surge of 2021–2022.

This is an encouraging first signal: our selected model appears to have successfully absorbed the underlying structure of US CPI returns, leaving behind only the unpredictable noise that no model can capture.

Residuals Distribution:

Having established that the residuals show no time-based pattern, we now examine their overall distribution, that is, we look at all the innovations together and ask: where do they cluster?

For a model to be considered unbiased, its residuals must be centered at zero. Intuitively, this means that across all months in our sample, the model's over-predictions and under-predictions should cancel each other out perfectly. If the mean of the residuals were significantly different from zero, it would indicate a systematic bias, for instance the model might consistently predict higher inflation than what was actually recorded. Such a bias would be a serious flaw, as it would cause every future forecast to be systematically off in the same direction.



Examining the residual distribution plots for all five models, we observe that in every case the distribution is symmetric and centered very close to zero. This holds for AR(1), AR(2), MA(1), ARMA(1,1), and the ARMA(2,1). There is no evidence of any directional bias in any of the candidate models, which is a reassuring result. However, it is worth noting that none of the models fit a perfect Normal distribution particularly well. All of them display a sharper central peak and heavier tails than a theoretical Gaussian curve would predict, suggesting that extreme inflation shocks occur more frequently than normality would assume. This is a common characteristic of macroeconomic data and does not invalidate our model, but it is an important limitation to keep in mind when interpreting the prediction intervals generated in the forecasting stage.

For our selected ARMA(2,1) model specifically, the mean of the residuals is effectively zero, confirming that the model neither systematically overestimates nor underestimates the monthly inflation rate. This unbiasedness remains a critical property: even if perfect normality is not achieved, the model still gives us a balanced and solid foundation from which to project the future path of US inflation.

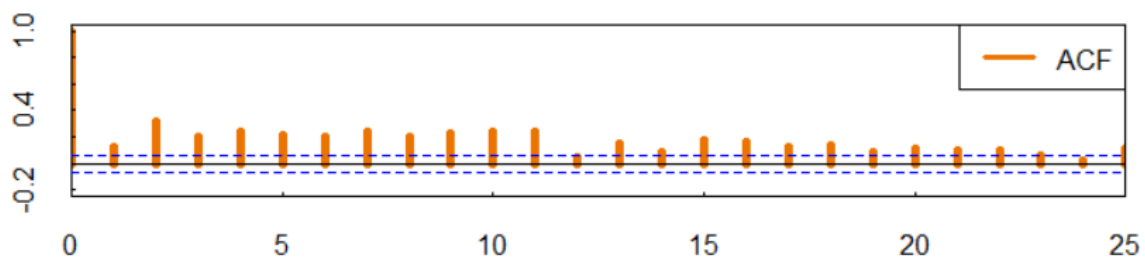
ACF / PACF of the Residuals

For our next step, we want to figure out how past values of our dataset influence present values. To do so, we use the autocorrelation and partial autocorrelation functions for our residuals.

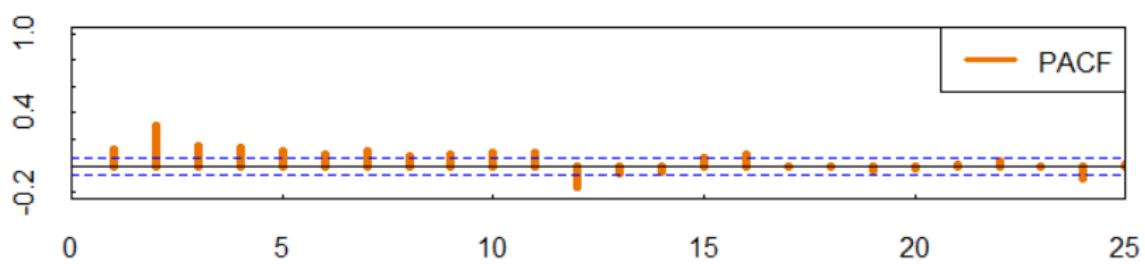
ACF measures the correlation between the residuals at the current time step and the residuals at previous time steps, capturing their direct and indirect effects whereas PACF does the same, but it strips away the influence of all the intervening lags.

These have been our results:

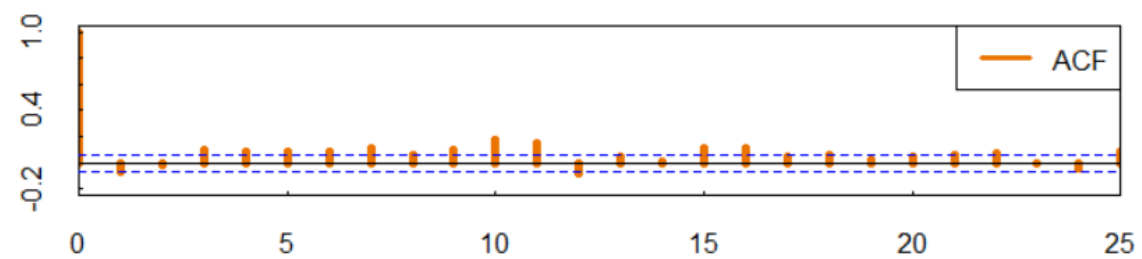
MA(1) : ACF of Residuals



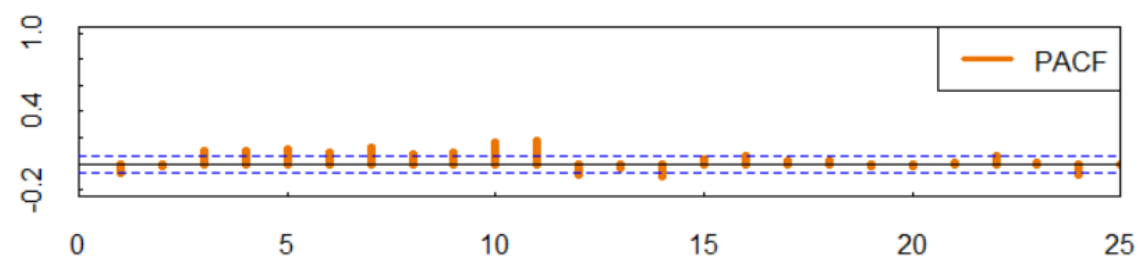
MA(1) : PACF of Residuals



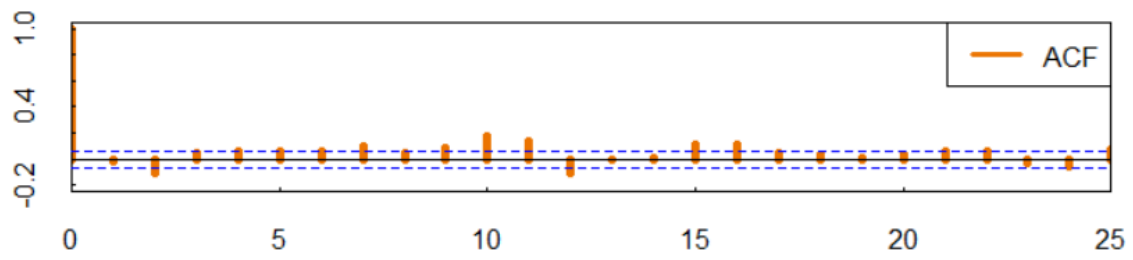
AR(1) : ACF of Residuals



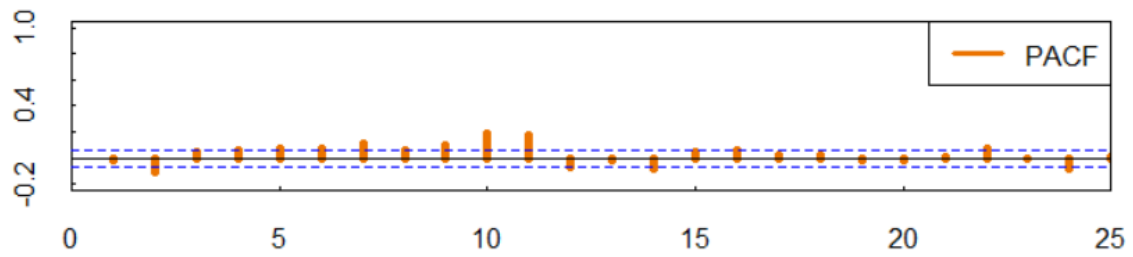
AR(1) : PACF of Residuals



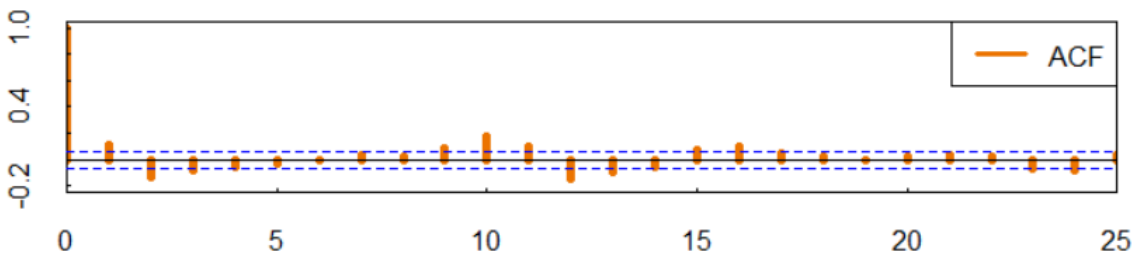
AR(2) : ACF of Residuals



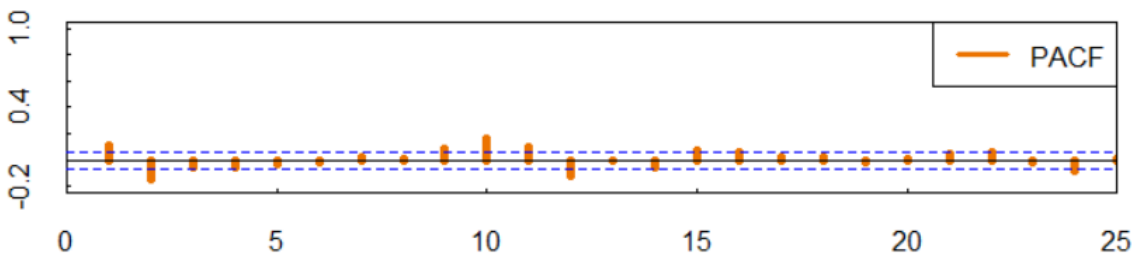
AR(2) : PACF of Residuals

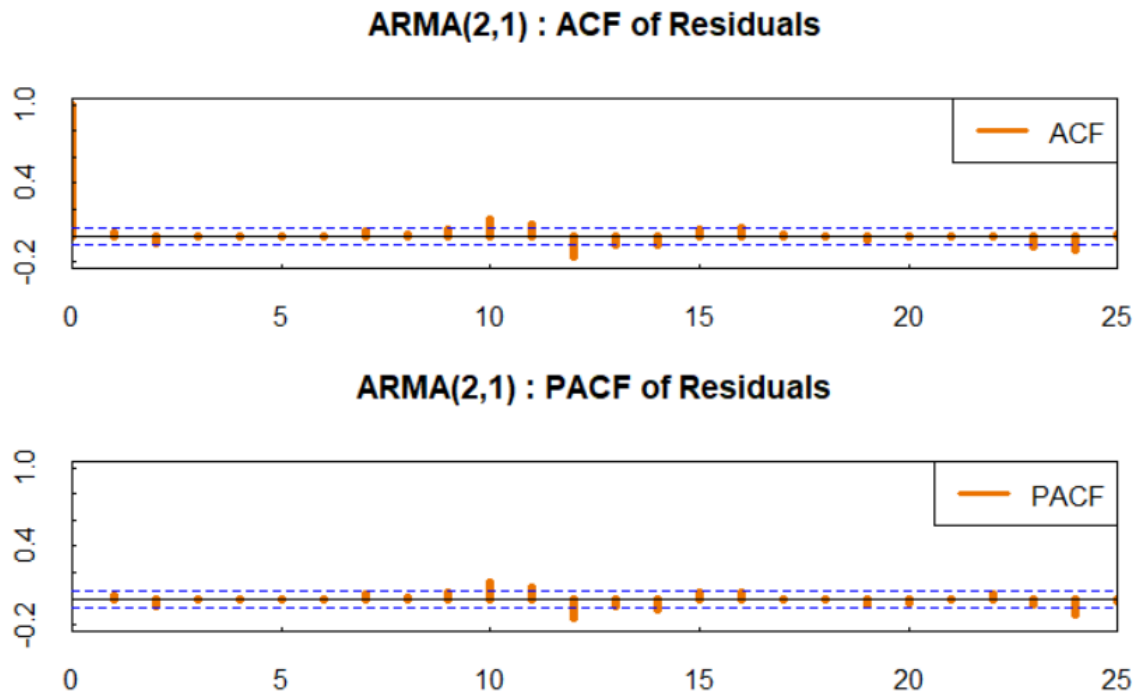


ARMA(1,1) : ACF of Residuals



ARMA(1,1) : PACF of Residuals





Based on our plots MA (1) shows some residual dependence and is the poorest fitting model of the group.

AR (1), AR (2) and ARMA (1,1) look cleaner but still don't fit at all and we could say ARMA (2,1) is the one that fits the best.

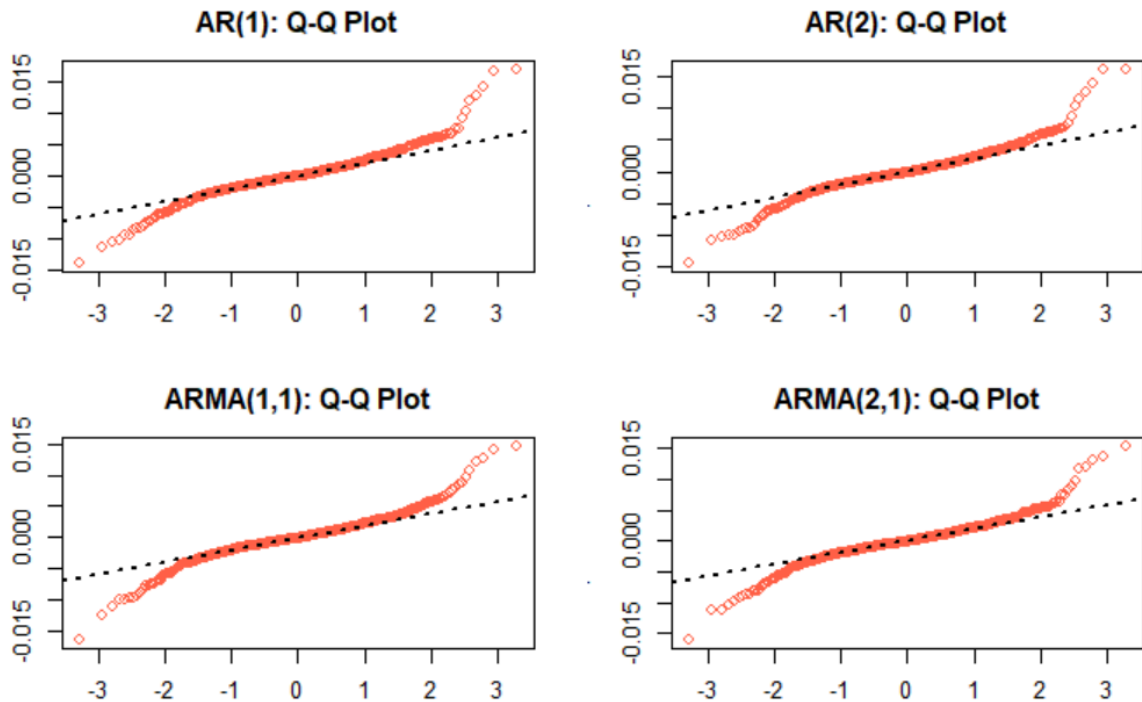
From these, we can highlight the persistent pattern at higher lags. We can also see that significant spikes above the blue line persist regardless of how many AR or MA terms we add.

QQ Plot (Quantile Quantile Plot)

To allow us to better see how much the tails of a distribution deviate from a Normal we use the QQ Plot (Quantile Quantile Plot).

We construct the graph by plotting the population quantiles of the normal distribution against the empirical quantiles of our data.

The more adjusted the orange points are to the 45 degree line, the more the data is congruent with a normal population.



Based on the Q-Q plots provided for the four models we can see that the residuals are not normally distributed. We see a distinct “S-curve” shape in each of these.

This indicates to us that our time series experiences outliers more frequently than the Normal would predict and that the models are underestimating the probability of very large positive and negative shocks.

Because it happens in all the models, it suggests to us that the problem is not how many lags we use but rather the assumption that the errors are normally distributed.

Residual JB Test (Jarque-Bera Test)

To verify the adequacy of an estimated model, we assume that the innovations behave as Gaussian white noise. This implies that the fitted residuals should approximately behave as a sample from a normal population. The Jarque-Bera test can be used to formally test the null hypothesis that the residuals are normal.

	AR(1)	AR(2)	ARMA(1,1)	ARMA(2,1)
Jarque-Bera (p-val)	0.000	0.000	0.000	0
Ljung-Box (p-val)	0.841	0.202	0.225	0

Upon analyzing the results table, the p-values for the AR(1), AR(2), ARMA(1,1), and ARMA(2,1) models are all exactly 0.000. Consequently, we strongly reject the null hypothesis in all cases. Mathematically, this indicates that the residuals do not follow a normal distribution, confirming the presence of thicker tails and a higher frequency of extreme values, as previously detected visually in the Q-Q plot.

Ljung-Box Test

A second fundamental requirement in the diagnostic checking phase is that the correlogram of the fitted residuals must not exhibit significant autocorrelation; that is, it should be clean. The Ljung-Box test can be used to test the null hypothesis that the residuals are not autocorrelated. The results show that the AR(1), AR(2), and ARMA(1,1) models yield p-values of 0.841, 0.202, and 0.225, respectively. For these three models, we fail to reject the null hypothesis, confirming that their residuals are indeed white noise.

However, the ARMA(2,1) model yields a p-value of 0.000, indicating the rejection of the null hypothesis and demonstrating that its residuals retain autocorrelation. The theoretical rule for model selection dictates that one must first choose among models which have good residual diagnostics, and then pick up the most parsimonious model in this set. Therefore, the ARMA(2,1) model should not be selected as the final optimal model, as it fails to pass this essential diagnostic check.

MSE (Mean Squared Error)

In the final forecasting stage, model performance is evaluated assuming the use of a loss function. Specifically, we assume that the loss function is quadratic, defined mathematically as: $L(y_{T+h}, \hat{y}_{T+h}) = (y_{T+h} - \hat{y}_{T+h})^2$. The objective of calculating the Mean Squared Error (MSE) is to measure the out-of-sample forecasting performance.

Among the four specifications tested, the ARMA(2,1) model provided the most precise fit, yielding a Root Mean Squared Error of 0.002734. This result suggests that inflation dynamics are best explained by a combination of two autoregressive lags and a single moving average term, which effectively balances the persistence of historical values with the decay of past random shocks.

[1] "--- Model Accuracy Comparison (MSE) ---"		
	Model	MSE
1	AR(1)	7.982811e-06
2	AR(2)	7.977447e-06
3	ARMA(1,1)	8.123721e-06
4	ARMA(2,1)	7.475681e-06

FORECASTING ANALYSIS

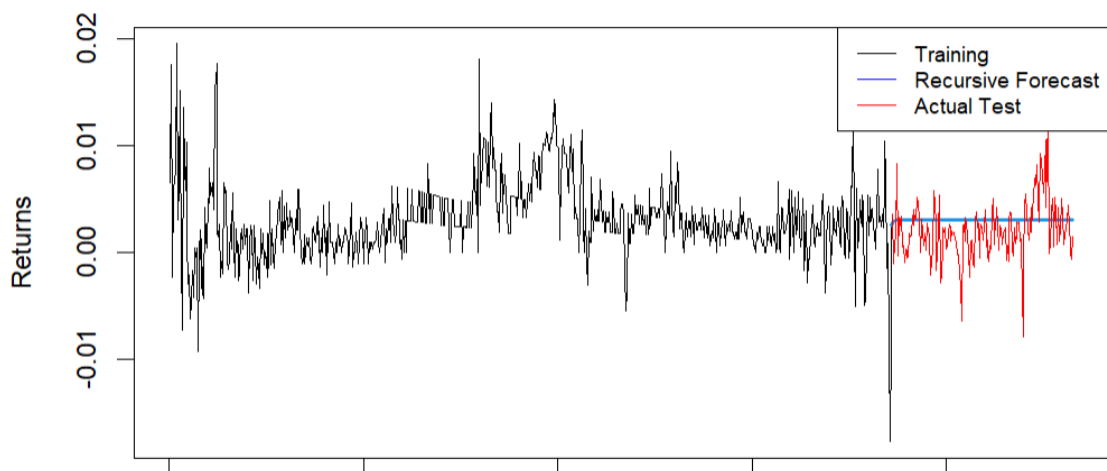
By utilizing an eighty-twenty split to partition the data into training and testing sets, we were able to evaluate the models' performance on unseen data, ensuring that the observed predictive power was not merely a result of over-fitting.

We will start with the recursive forecasting approach, which conditions predictions on the state of the system at time t and uses all the available information until that time. It allows the autoregressive and moving average components to project the series toward its long-term equilibrium, demonstrating that inflation shocks possess a quantifiable memory.

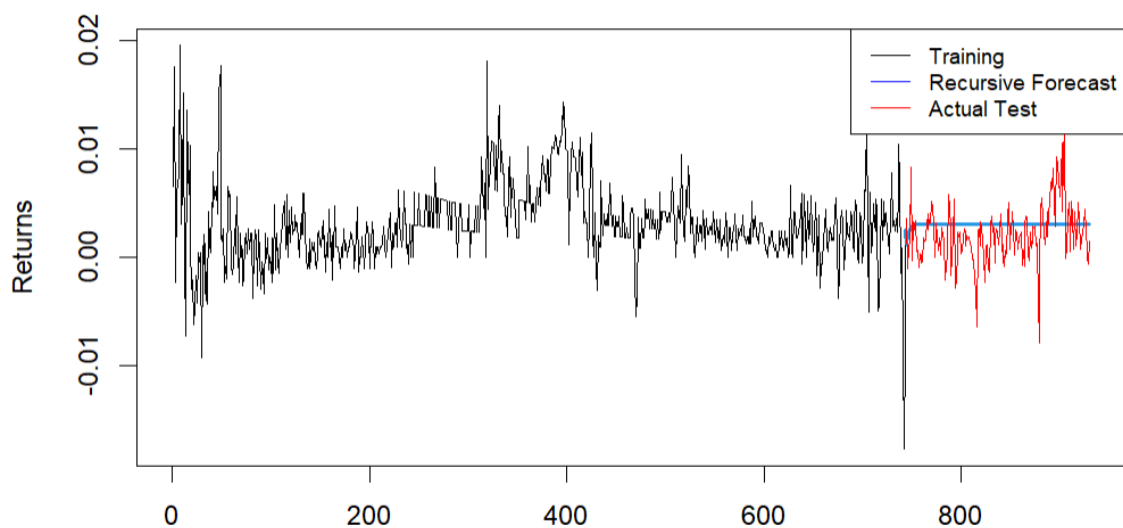
As expected, we can see that the forecast for the AR(p) models goes to zero after lag p, due to their definition. By contrast, the models including a moving average like ARMA(p,q) models show an exponential return to the mean.

The models do not capture the high-frequency volatility or exogenous shocks visible in the actual test data. This discrepancy between the smooth forecast trajectory and the noisy realized returns highlights the inherent difficulty in timing inflation fluctuations, yet it confirms that the ARMA(2,1) specification provides the most reliable statistical anchor for long-term expectation setting in this dataset.

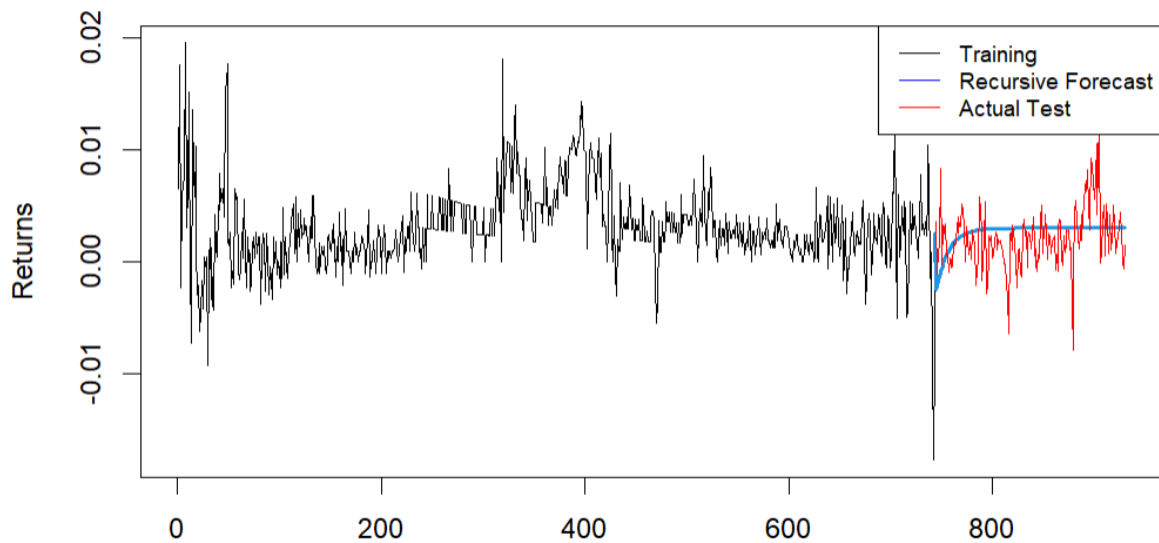
AR(1) Recursive Forecast (80/20 Split)



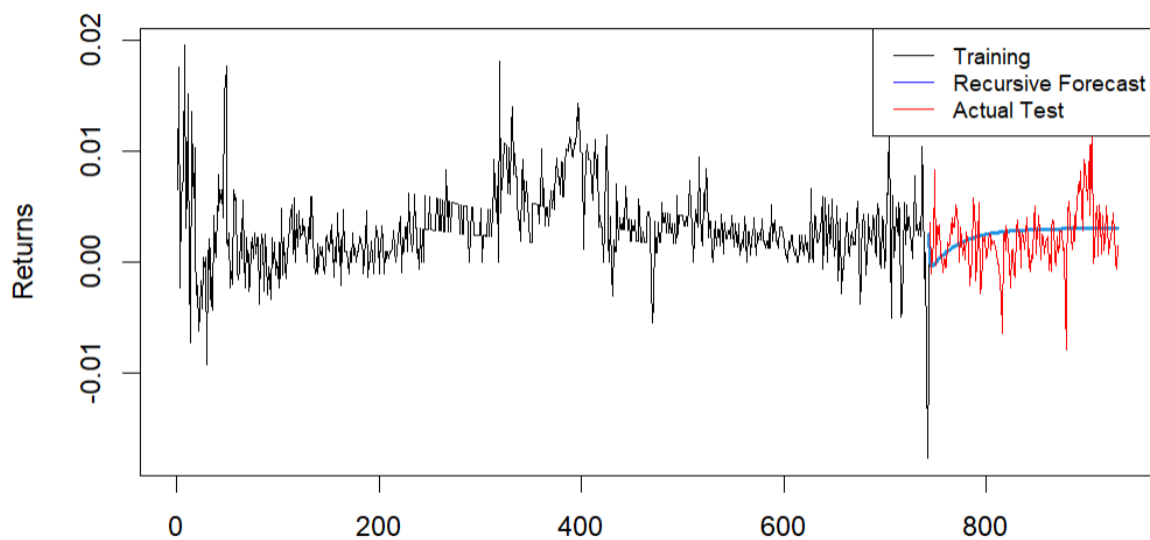
AR(2) Recursive Forecast (80/20 Split)



ARMA(1,1) Recursive Forecast (80/20 Split)



ARMA(2,1) Recursive Forecast (80/20 Split)

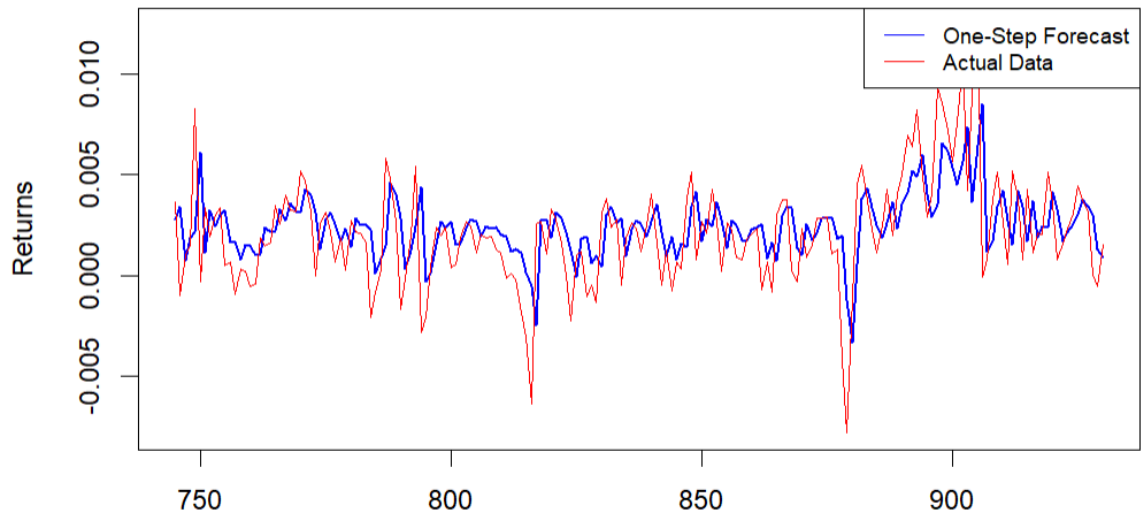


In the next part of the project we show the 1 lag ahead forecast, using all information until time t . This is very similar to the way we calculated the residuals in the first part of the project.

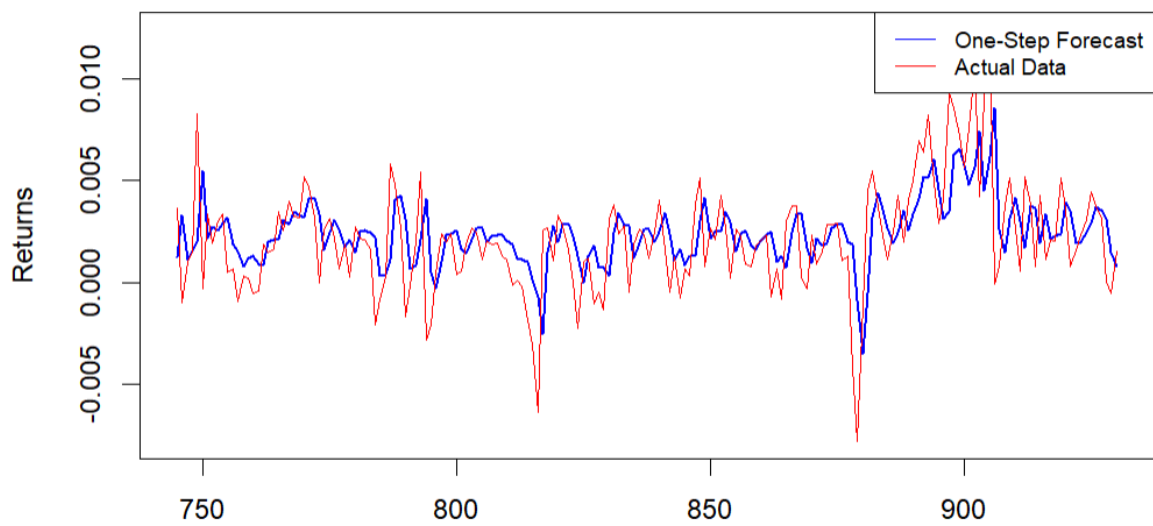
We can observe that all the models fail to forecast extreme shocks and stay in a more conservative position closer to the mean.

Although all models seem to have a good feed, ARMA(2,1) seems to be slightly better. We can prove that by looking at the MSE of the four models, where we can see that ARMA(2,1) has the smallest error.

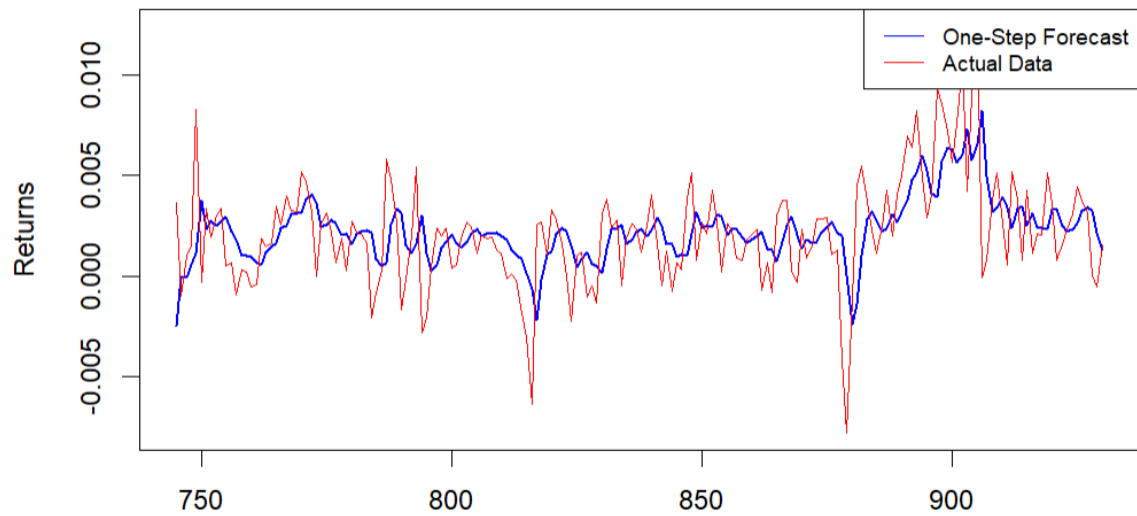
AR(1) One-Step-Ahead Rolling Forecast



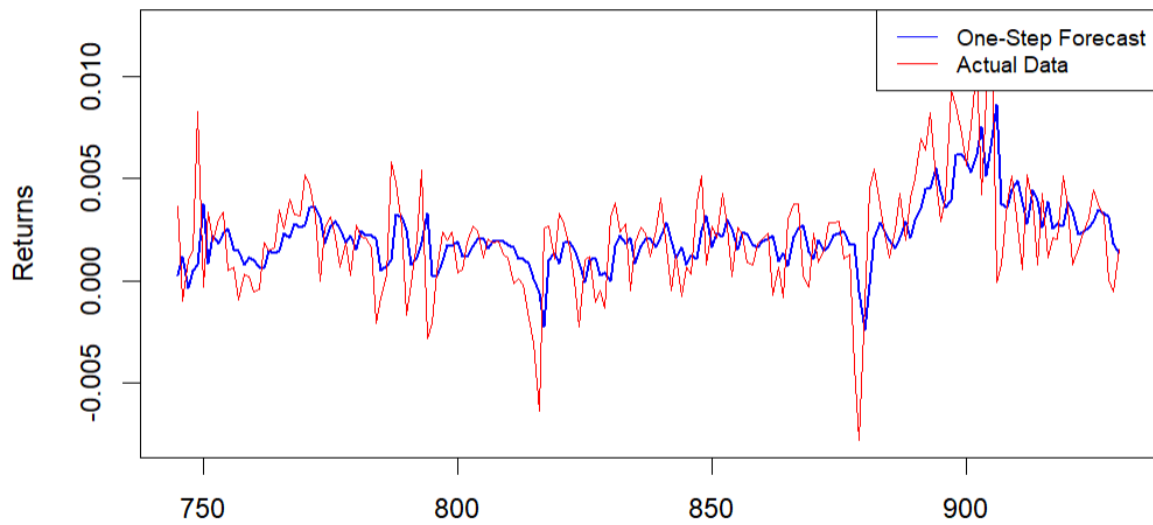
AR(2) One-Step-Ahead Rolling Forecast



ARMA(1,1) One-Step-Ahead Rolling Forecast



ARMA(2,1) One-Step-Ahead Rolling Forecast



	Model	MSE
1	AR(1)	5.375145e-06
2	AR(2)	5.477857e-06
3	ARMA(1,1)	5.626896e-06
4	ARMA(2,1)	5.158871e-06

CONCLUSION

The aim of this project has been to conduct an exhaustive analysis and predict the dynamics of the CPI in the United States to understand how this variable evolves over time. To do this, we applied the full forecasting procedure, from identification, estimation, and diagnostics through selection and prediction.

For doing that, we compared five candidate models (AR(1), AR(2), MA(1), ARMA(1,1), and ARMA(2,1)) by estimating their parameters using maximum likelihood, diagnosing their residuals with the Ljung-Box test, and testing their out-of-sample predictive performance using the mean squared error (RMSE).

Model selection:

To choose which model best represents the dynamics of the US CPI, we evaluated five different models: AR(1), AR(2), MA(1), MA(2), ARMA(1,1) and ARMA(2,1), running various tests and diagnostics:

In the first model selection phase, we estimated the parameters using the Log-Likelihood and applied the AIC and BIC criteria, which penalise the inclusion of unnecessary parameters and overparameterisation, and we aim to minimise them. The ARMA(2,1) model stands out as the best option in this regard because it has the highest Log-Likelihood value = 4182,13, and the lowest one in the AIC and BIC Tests, whose values are: -8,985 and -8,964 respectively.

Secondly, by doing the residuals diagnostics checks to ensure they mimic White Noise, we maintain our initial hypothesis that the ARMA(2,1) model is the best, as it produces the most well-distributed and stable residuals.

However, the Ljung-Box Test indicates that the simpler models, such as: AR(1) = 0,841, AR(2) = 0,202, ARMA(1,1) = 0,225; have p-values $> 0,05$ and therefore do not reject the null hypothesis, confirming that their residuals are not autocorrelated. In contrast, the ARMA(2,1) model obtains a p-value = $0,00 < 0,05$. This is sufficient to reject the null hypothesis and serves to confirm that some autocorrelation remains in its residuals. The theory tells us that this is sufficient reason to discard what was until now the best-chosen model, and that we should choose a simpler one, such as the ARMA(1,1). However, the selection of a model is not based solely on in-sample fit but also on its ability to make out-of-sample predictions.

And finally, we have done the forecasting analysis by splitting the data into two groups, 80% for training and 20% for testing. The ARMA(2,1) model once again proves to be the best model for predicting future data, setting long-term expectations and explaining the dynamics of the IPC. This is all thanks to achieving the lowest RMSE value of 0,002734.

Synthesizing these findings, we conclude that the best model is the **ARMA(2,1)**, despite the penalty in the Ljung-Box Test its ability to set long-term expectations and accurately project future out of sample data makes it the definitive choice for explaining US inflation dynamics.

Using various diagnostic tests, we were able to observe that the residuals exhibit thicker tails than a standard normal distribution. This characteristic suggests the presence of “volatility clustering”, a phenomenon in which periods of large inflationary shocks tend to be followed by further large shocks, and periods of calm by further calm.

A future analysis to expand on the research would be to evaluate ARCH and GARCH models, as this would likely resolve the diagnostic checking issues with the residuals, taking into account the thicker tails observed in our distribution, and provide a deeper understanding of the uncertainty and risk associated with inflation dynamics in the US.

ARMA(2,1) PARAMETERS for predicting the coming months.

- $c = 0$
- $\phi_1 = 1,26$
- $\phi_2 = -0,29$
- $\theta_1 = -0,85$